Tanisha Prasad

Embedded Firmware Engineer

RESUME_1907

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Total Experience: 10 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 10 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: ARM Cortex-M, I2C, UART, C++, C, STM32, Embedded Linux, Firmware Debugging, Oscilloscope, SPI, CAN, FreeRTOS

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Lead Embedded Firmware Engineer | Urbanaut Engineering | Jan 2023 - Present

- Lowered field-reset incidents by 21% with watchdog recovery and diagnostics
- Reduced embedded task jitter by 20% using interrupt priority and queue tuning
- Validated 202 hardware test vectors across protocol and timing corner cases
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Embedded Firmware Engineer | LuminaForge Technologies | Jan 2019 - Dec 2022

- Validated 647 hardware test vectors across protocol and timing corner cases
- Improved battery runtime by 27% with peripheral and sleep-state optimization

Associate Embedded Firmware Engineer | NovaPeak Engineering | Jul 2016 - Dec 2018

- Lowered field-reset incidents by 22% with watchdog recovery and diagnostics
- Reduced embedded task jitter by 36% using interrupt priority and queue tuning

PROJECTS

Motor Control Interface | STM32, SPI, C, C++

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | ARM Cortex-M, FreeRTOS, Firmware Debugging, CAN

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Crescent Valley Technical University, 2016

Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

Embedded C Standards (2020)